

Algebra 1
Unit 8
Lesson 6 Practice

Name **Answer Key** _____
Date _____
Period _____

Re-write each expression into vertex form by completing the square.

1. $4t^2 - 16t + 12$
 $4(t^2 - 4t + 4) + 12 - 16$
 $4(t - 2)^2 - 4$

2. $-2x^2 + 8x - 18$
 $-2(x^2 - 4x + 4) - 18 + 8$
 $-2(x - 2)^2 - 10$

3. $3r^2 - 18r - 12$
 $3(r^2 - 6r + 9) - 12 - 27$
 $3(r - 3)^2 - 39$

4. $-5h^2 - 20h + 3$
 $-5(h^2 + 4h + 4) + 3 + 20$
 $-5(h + 2)^2 + 23$

5. $3y^2 - 24y - 4$
 $3(y^2 - 8x + 16) - 4 - 48$
 $3(y - 4)^2 - 52$

6. $-2u^2 + 12u + 7$
 $-2(u^2 - 6u + 9) + 7 + 18$
 $-2(u - 3)^2 + 25$

7. $10n^2 + 40n + 68$
 $10(n^2 + 4n + 4) + 68 - 40$
 $10(n + 2)^2 + 28$

8. $-3d^2 - 18d + 4$
 $-3(d^2 + 6d + 9) + 4 + 27$
 $-3(d + 3)^2 + 31$

9. $4y^2 + 48y - 15$
 $4(y^2 + 12y + 36) - 15 - 144$
 $4(y + 6)^2 - 159$

10. $9c^2 - 18c - 23$
 $9(c^2 - 2c + 1) - 23 - 9$
 $9(c - 1)^2 - 32$